

Potential Flow Over Airfoils

Airfoil nomenclature

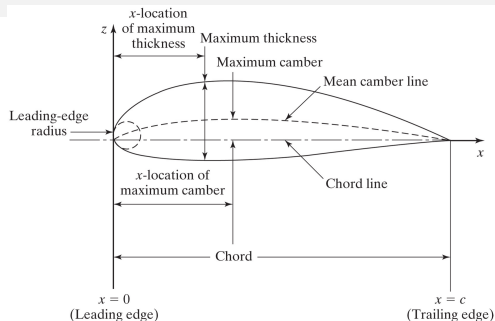


Figure 4.1: Airfoil nomenclature, Fig. 5.6 from AE-BC

- ▶ **chord (c) /chord line** - Straight line connecting leading edge to trailing edge
- ▶ **mean camber line** - locus of points midway between upper and lower surface
- ▶ **thickness(x)** - upper surface - lower surface
- ▶ **camber** - location of maximum camber - for symmetric airfoil = 0

Forces on an airfoil

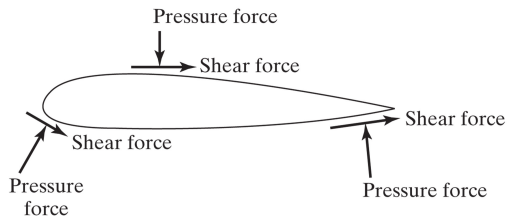


Figure 4.2: Forces on airfoil, Fig. 5.2 from AE-BC

- ▶ pressure $p(\vec{s})$ - this is the static pressure and always acts **normal** to the surface
- ▶ wall shear stress (force) or skin friction τ_w - always acts **tangential** to the surface

Forces on an airplane

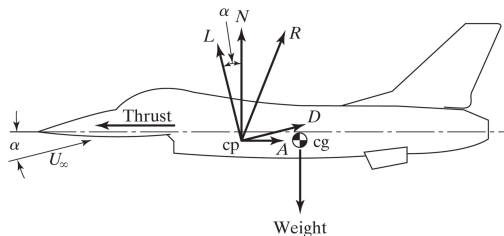


Figure 4.3: Forces on an airplane, Fig. 5.3 from AE-BC

- ▶ R = resultant force
- ▶ L = lift force - correspond force on an airfoil (2-D) l (per unit span)
- ▶ N = normal force
- ▶ A = axial force
- ▶ D = drag force - correspond force on an airfoil (2-D) d (per unit span)

Forces on an airplane

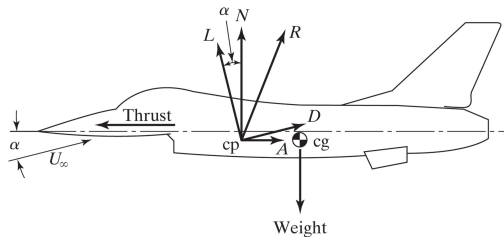


Figure 4.4: Forces on an airplane, Fig. 5.3 from AE-BC

$$\begin{bmatrix} D \\ L \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} A \\ N \end{bmatrix} \quad (4.1)$$

Moments on an airplane

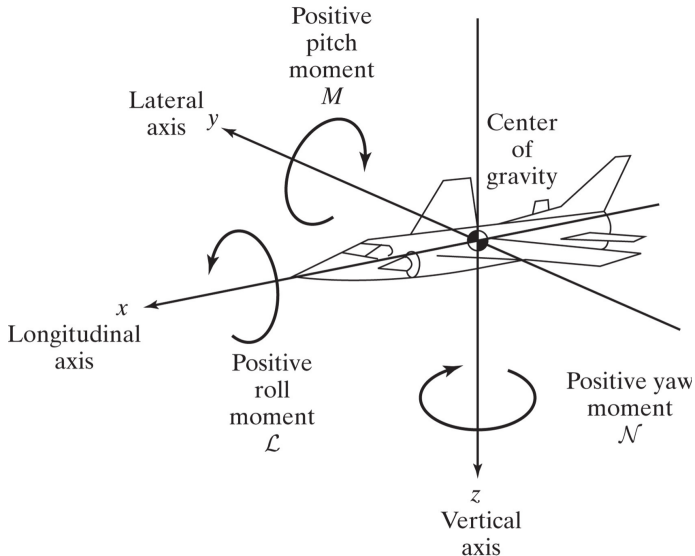


Figure 4.5: Moments on an airplane, Fig. 5.5 from AE-BC

Moments on an airfoil

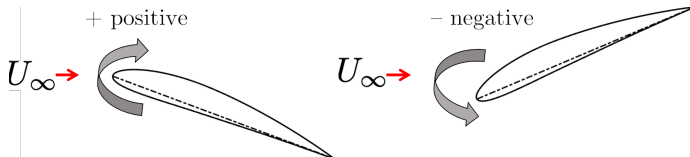


Figure 4.6: Sign convention for moments on an airfoil

- ▶ Define around different points - usually leading edge m_0 or quarter chord ($c/4$) $m_{c/4}$
- ▶ positive moment - pitch up , increases α
- ▶ negative moment - pitch down , decreases α

Non-Dimensional Numbers - wing

- Lift coefficient -

$$C_L = \frac{L}{q_\infty S} \quad (4.2)$$

- ▶ Drag coefficient -

$$C_D = \frac{D}{q_\infty S} \quad (4.3)$$

- ▶ Moment coefficient

$$C_M = \frac{M}{q_\infty S c} \quad (4.4)$$

$$q_{\infty} = \frac{1}{2} \rho_{\infty} U_{\infty}^2 \rightarrow \text{dynamic pressure} \quad (4.5)$$

$$S = b \times c \rightarrow \text{planform area, } b \rightarrow \text{width of wing} \quad (4.6)$$

Non-Dimensional Numbers - airfoil

- ▶ Lift coefficient -

$$C_l = \frac{l}{q_\infty c} \quad (4.7)$$

- ▶ Drag coefficient -

$$C_d = \frac{d}{q_\infty c} \quad (4.8)$$

- ▶ Moment coefficient

$$C_m = \frac{m}{q_\infty c^2} \quad (4.9)$$

Other Non-Dimensional Numbers

- ▶ Pressure coefficient -

$$C_p = \frac{p - p_\infty}{q_\infty} \quad (4.10)$$

- ▶ Skin friction coefficient -

$$C_f = \frac{\tau_w}{q_\infty} \quad (4.11)$$

Airfoil Nomenclature - NACA 4 digit

NACA 2412

$$2 \rightarrow \text{max. camber in } \%c = 100m \quad (4.12)$$

$$4 \rightarrow \text{position of max. camber in } c/10 = (10p) \quad (4.13)$$

$$12 \rightarrow \text{max. thickness in } \%c = 100t \quad (4.14)$$

$$z_c = \begin{cases} \frac{x}{p^2} \left(2p - \frac{x}{c} \right), & 0 \leq x \leq pc \\ \frac{c-x}{(1-p)^2} \left(1 + \frac{x}{c} - 2p \right), & pc \leq x \leq c \end{cases} \quad (4.15)$$

$$z_t = 5tc \left[0.2969 \sqrt{\frac{x}{c}} + (-0.1260) \left(\frac{x}{c} \right) + (-0.3516) \left(\frac{x}{c} \right)^2 \right. \\ \left. + 0.2843 \left(\frac{x}{c} \right)^3 + (-0.1015) \left(\frac{x}{c} \right)^4 \right] \quad (4.16)$$

NACA 4 digit

$$\theta = \tan^{-1} \frac{dz_c}{dx} \quad (4.17)$$

$$x_U = x - z_t \sin \theta, \quad z_U = z_c + z_t \cos \theta, \quad (4.18)$$

$$x_L = x + z_t \sin \theta, \quad z_L = z_c - z_t \cos \theta, \quad (4.19)$$

Equations are available in Abbott and Doenhoff “Theory of Wing Sections” or like me you can shamelessly copy it from Wikipedia - the ones on there are correct and the explanation is good.

https://en.wikipedia.org/wiki/NACA_airfoil

Refer to Abbott and Doenhoff “Theory of Wing Sections” for 5-digit and 6-digit sections.

Center of Pressure

Center of pressure: Location of the single force - resultant R (per unit span) (see Fig. 4.10) that produces the same moment as the distributed load (from p and τ_w , see Fig. 4.2), on the airfoil

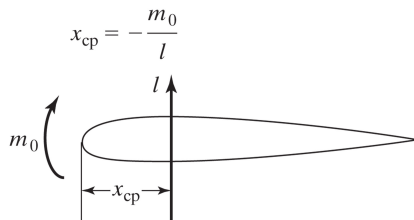


Figure 4.7: center of pressure, Fig 6.7 from AE-BC

$$m_0 = -lx_{cp} \quad (4.20)$$

$$x_{cp} = \frac{-m_0}{l} \quad (4.21)$$

Center of Pressure

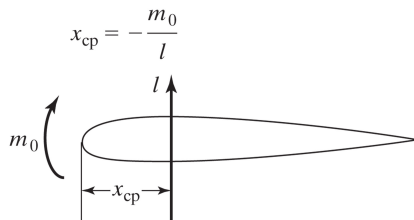


Figure 4.8: center of pressure, Fig 6.7 from AE-BC

$$x_{cp} = \frac{-m_0}{l}$$

Note, here it is assumed that at small angles of attack (more on this shortly) $l \approx$ normal force (N per unit span in Fig. 4.10)

For thin symmetric airfoils $x_{cp} = c/4$

Small angle assumption

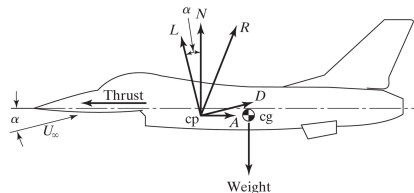


Figure 4.9: Forces on an airplane, Fig. 5.3 from AE-BC

If α is small $\rightarrow \sin \alpha \approx \alpha$, note – in radians, (4.22)

$\cos \alpha \approx 1$ (4.23)

$\tan \alpha \approx \alpha$, note – in radians (4.24)

$\alpha \times \text{something small} \approx 0$ (4.25)

$L \approx N$ (4.26)

Transfer of moments

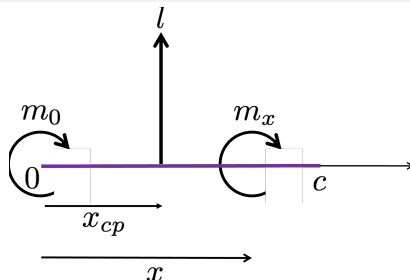


Figure 4.10: Transfer for moments

Moments can be transferred between points

$$m_x = xl + m_0 \quad (4.27)$$

where x is the distance between the point about which the moment is calculated and the leading edge. For example $m_{c/4} = c/4l + m_0$.

Aerodynamic center

In general, the center of pressure for a cambered airfoil is

1. a function of angle of attack α
2. Note - $C_l = f(\alpha, \dots)$ - more on this soon.
3. Thus from a stability and control perspective x_{cp} is a difficult to use.

Aerodynamic center: x_{ac} The point at which the moment coefficient is invariant with Lift or C_l or C_L or α

$$\text{For an airfoil at } x_{ac}: \frac{dC_{m,ac}}{dC_l} = \frac{dC_{m,ac}}{d\alpha} = 0 \quad (4.28)$$

For thin airfoils x_{ac} close to $c/4$

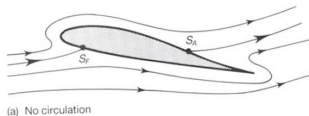
Potential flow recap

From what we learned about potential flow basics we can conclude a few things

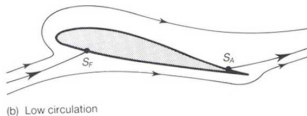
- ▶ If we use only uniform flows, sources, sinks & doublets to generate flow around bodies we cannot generate lifting flows .
- ▶ But we can create flow over bodies with finite thickness.
- ▶ However, to get lift we need to add a vortex to the flow. Note - a vortex has infinite vorticity at its center. Everywhere but exactly at the vortex center, the flow is irrotational.
- ▶ To summarize uniform flow + source(s)/sink(s)/doublet(s) gives us thickness and we add vortex (or vortices) to get lift.

Let's for the moment assume that by using a combination of sources/sinks/doublets in a uniform flow we can generate non-lifting flow over an airfoil. Now, we add a vortex with strength Γ to this flow field. Based on the magnitude of Γ we choose we can get a few flow scenarios.

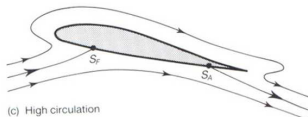
Kutta Condition



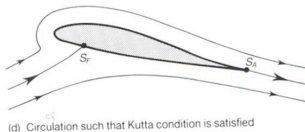
$$\Gamma_a = 0$$



$$0 = \Gamma_a < \Gamma_b < \Gamma_c$$



$$\Gamma_b < \Gamma_c < \Gamma_d$$



$\Gamma \rightarrow$ Kutta condition

Figure 4.11: Potential flow over an airfoil, adapted from AE-HCCV

Kutta Condition

For potential flow over an airfoil there is only one Γ or circulation that moves the rear stagnation point to the trailing edge. We arrive at this conclusion by visualizing real flow over an airfoil. As long as there is no flow separation flow leaves the airfoil trailing edge smoothly. Note, in a potential flow, the streamlines can flow over the airfoil smoothly only when the rear stagnation point is at the trailing edge. In a real flow what causes the circulation that keeps the streamlines flowing smoothly over the airfoil when there is no flow separation? Viscosity. Hence, adding a vortex to a potential flow is one way of mimicking the effect of viscosity. Note, at the center of a vortex the vorticity is infinite (rotational - viscous).

When we add a vortex to potential flow over an airfoil is an airfoil spinning? NO! But if we subtract out the uniform flow (free stream) the flow field looks very interesting.

Circulation theory of lift

show video

Kutta-Joukowski Theorem

From flow over a rotating cylinder, we derived lift per unit span to be,

$$l = \rho_{\infty} U_{\infty} \Gamma \quad (4.29)$$

This relationship is true for ALL potential flows. This can be easily verified.

Kutta condition for vortex sheet

So far we have considered point sources, sinks, doublets and vortices. Let us consider a distributed vorticity or a vortex sheet.

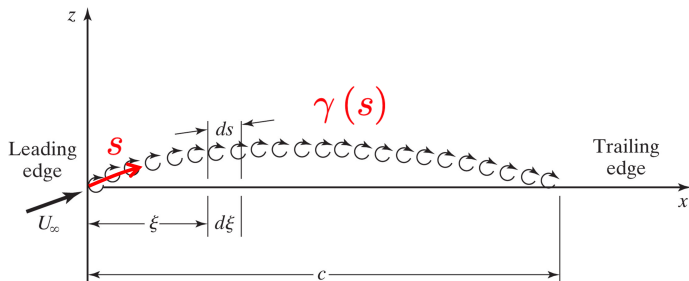


Figure 4.12: Mean camber line represented as a vortex sheet, adapted Fig 6.4 from AE-BC

Vortex strength per unit length is $\rightarrow \gamma(s)$ Note:

$$\Gamma = \int_0^c \gamma(s) ds \quad (\text{TEq. (6.1)}) \quad (4.30)$$

$$\text{Kutta condition} \rightarrow \gamma(c) = \gamma(TE) = 0 \quad (4.31)$$

Thin airfoil theory

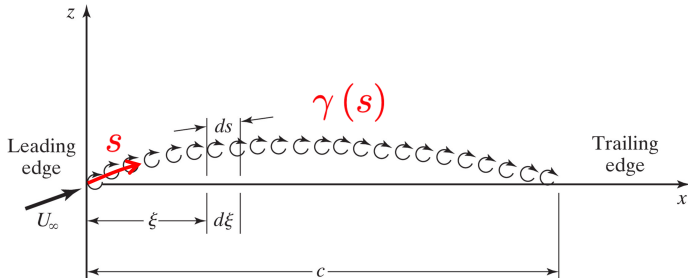


Figure 4.13: Mean camber line represented as a vortex sheet, adapted Fig 6.4 from AE-BC

Thin Airfoil Theory: Shrink vorticity (circulation) caused by airfoil to a **vortex sheet** of strength $\gamma(s)$ Place vortex sheet $\gamma(s)$ on **camber line** and solve for $\gamma(s)$ such that

i **camber line is a streamline** (no flow through it)

ii $\gamma(TE) = \gamma(c) = 0 \rightarrow$ Kutta condition

Thin airfoil theory

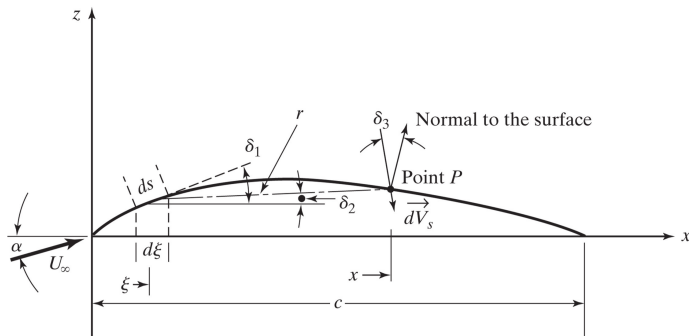


Figure 4.14: TAT geometry - no flow through camber line, Fig 6.5 from AE-BC
The elemental portion of the vortex sheet ds induces a velocity dV_s at point P . The normal component of this velocity is

$$dV_{s,n} = -\frac{\gamma ds \cos \delta_3}{2\pi r} \quad (4.32)$$

Thin airfoil theory

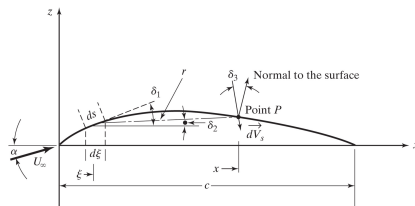


Figure 4.15: TAT geometry - no flow through camber line, Fig 6.5 from AE-BC

This is the contribution from the small piece ds . Therefore, the contribution from the entire sheet at this point P located arbitrarily at a distance x from the leading edge is

$$V_{s,n}(x) = -\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi) \cos \delta_2 \cos \delta_3 d\xi}{(x-\xi) \cos \delta_1} \quad (\text{TEq. (6.2)}) \quad (4.33)$$

Since x was chosen arbitrarily this is the expression for $V_{s,n}$ at any x i.e., point on the mean camber line.

Thin airfoil theory

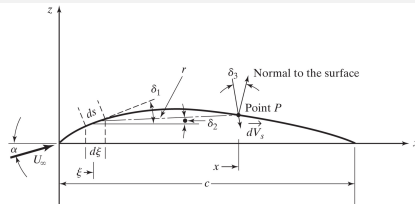


Figure 4.16: TAT geometry - no flow through camber line, Fig 6.5 from AE-BC

Similarly, the free-stream velocity U_∞ also contributes to the normal velocity at the P . Therefore, the normal component of the free-stream velocity is,

$$U_{\infty,n}(x) = U_\infty \sin(\alpha - \delta_p) \quad \& \quad \delta_p = \tan^{-1} \frac{dz}{dx} \quad (4.34)$$

Here, dz/dx is the slope of the mean camber line. Finally,

$$U_{\infty,n}(x) = U_\infty \left(\alpha - \tan^{-1} \frac{dz}{dx} \right) \quad (\text{TEq. (6.3)}) \quad (4.35)$$

Thin airfoil theory

For there to be no flow through the mean camber line i.e., $\vec{V} \cdot \hat{n} = 0$ or the mean camber line is a streamline, then $V_{s,n}$ (eq. (4.33)) and $U_{\infty,n}$ (eq. (4.35)) have to be equal and opposite. Therefore, the statement for $\vec{V} \cdot \hat{n} = 0$ on the mean camber line is

$$\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi) \cos \delta_2 \cos \delta_3 d\xi}{(x - \xi) \cos \delta_1} = U_{\infty} \left(\alpha - \tan^{-1} \frac{dz}{dx} \right) \quad (\text{TEq. (6.4)}) \quad (4.36)$$

Finally, all angles are “small” angles by definition as the airfoil is “thin” (rule of thumb angles $< 20^\circ$). Using the relationships in slide 145 we get

$$\cos \delta_2 = \cos \delta_3 = \cos \delta_1 \approx 1 \quad (4.37)$$

$$\frac{dz}{dx} = \tan \delta_P \approx \delta_P \rightarrow \tan^{-1} \frac{dz}{dx} = \frac{dz}{dx} \quad (4.38)$$

Fundamental Eqn. of TAT

Hence, the mathematical statement for no flow through the mean camber line becomes

$$\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi) d\xi}{(x-\xi)} = U_\infty \left(\alpha - \frac{dz}{dx} \right) \quad (\text{TEq. (6.5)}) \quad (4.39)$$

Knowns: $z(x)$, U_∞ & α

Unknown: $\gamma(\xi)$

This is the fundamental eq. of thin airfoil theory.

Fundamental Eqn. of TAT

$$\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi) d\xi}{(x-\xi)} = U_\infty \left(\alpha - \frac{dz}{dx} \right) \quad (\text{TEq. (6.5)}) \quad (4.40)$$

This is an integral equation and not trivial to solve. Often it is easy to work in trigonometric functions. Hence, we carry out the following transformation.

$$\text{Let, } \xi = \frac{c}{2} (1 - \cos \theta), \quad 0(\text{LE}) \leq \theta \leq \pi(\text{TE}) \quad (4.41)$$

$$x = \frac{c}{2} (1 - \cos \theta_0), \quad 0(\text{LE}) \leq \theta_0 \leq \pi(\text{TE}) \quad (4.42)$$

$$d\xi = \frac{c}{2} \sin \theta d\theta \quad (4.43)$$

$$\theta_0 = 0, \rightarrow x = 0(\text{LE}) \quad (4.44)$$

$$\theta_0 = \pi, \rightarrow x = c(\text{TE}) \quad (4.45)$$

Fundamental Eqn. of TAT

Finally, the fundamental eqn. of thin airfoil theory becomes,

$$\frac{1}{2\pi} \int_0^\pi \frac{\gamma(\theta) \sin \theta d\theta}{(\cos \theta - \cos \theta_0)} = U_\infty \left(\alpha - \frac{dz}{dx} \right) \quad (\text{TEq. (6.19)}) \quad (4.46)$$

Knowns: $z(x)$ (really $dz/dx(\theta)$), U_∞ & α

Unknown: $\gamma(\theta)$

General solution of fundamental eqn. of TAT

$$\frac{1}{2\pi} \int_0^\pi \frac{\gamma(\theta) \sin \theta}{(\cos \theta - \cos \theta_0)} d\theta = U_\infty \left(\alpha - \frac{dz_c}{dx} \right) \quad (4.47)$$

For a cambered airfoil $z(x) \neq 0$. We represent $dz(x)/dx$ as our old friend **Fourier cosine series**

$$\frac{dz}{dx} = (\alpha - A_0) + \sum_{n=1}^{\infty} A_n \cos(n\theta), \quad (\text{TEq. (6.22)}) \quad (4.48)$$

$$\text{here, } A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \left(\frac{dz}{dx} \right) d\theta \quad (\text{TEq. (6.23)}) \quad (4.49)$$

$$A_n = \frac{2}{\pi} \int_0^\pi \left(\frac{dz}{dx} \right) \cos(n\theta) d\theta \quad (\text{TEq. (6.24)}) \quad (4.50)$$

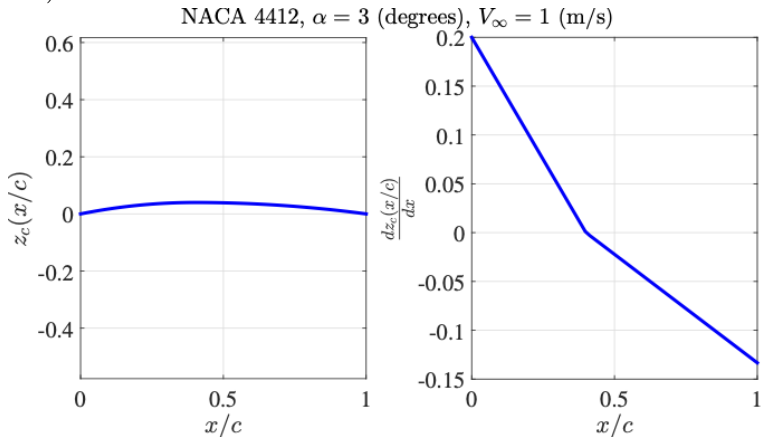
After some math magic, the solution:

$$\gamma(\theta) = 2U_\infty \left[A_0 \frac{1 + \cos \theta}{\sin \theta} + \sum_{n=1}^{\infty} A_n \sin(n\theta) \right] \quad (\text{TEq. (6.20)})$$

(4.51)

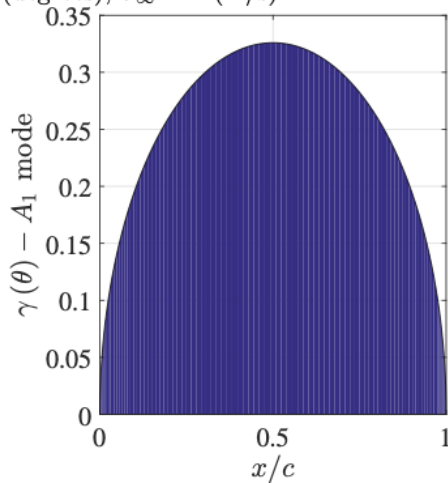
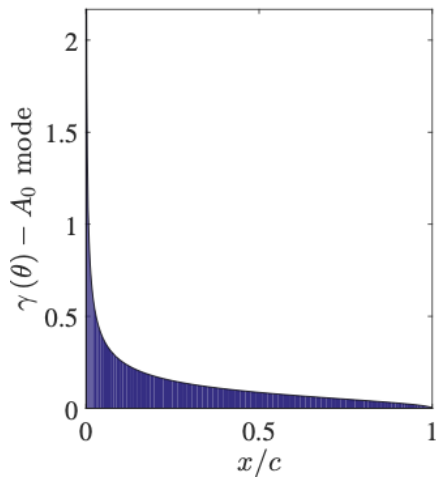
Solution for NACA 2412

Solved for NACA 4412 at $\alpha = 3$ degrees and $U_\infty \text{ ms}^{-1}$ (MATLAB code uploaded)



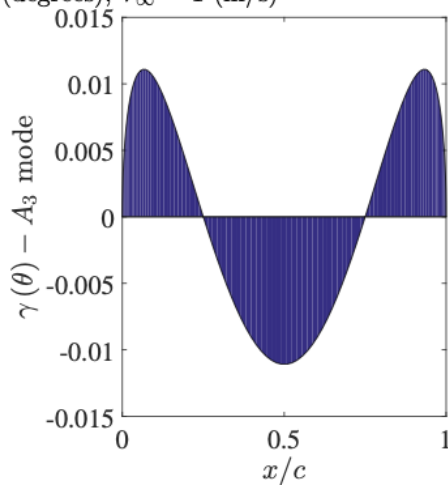
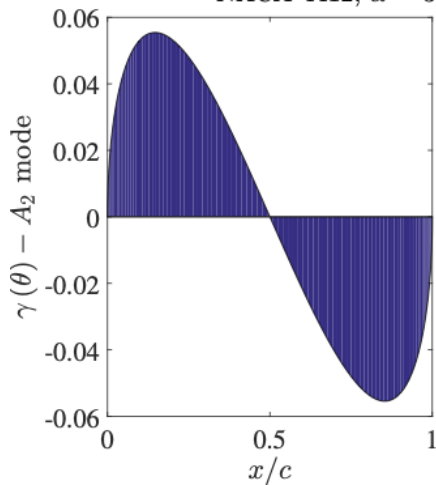
Solution for NACA 2412

NACA 4412, $\alpha = 3$ (degrees), $V_\infty = 1$ (m/s)

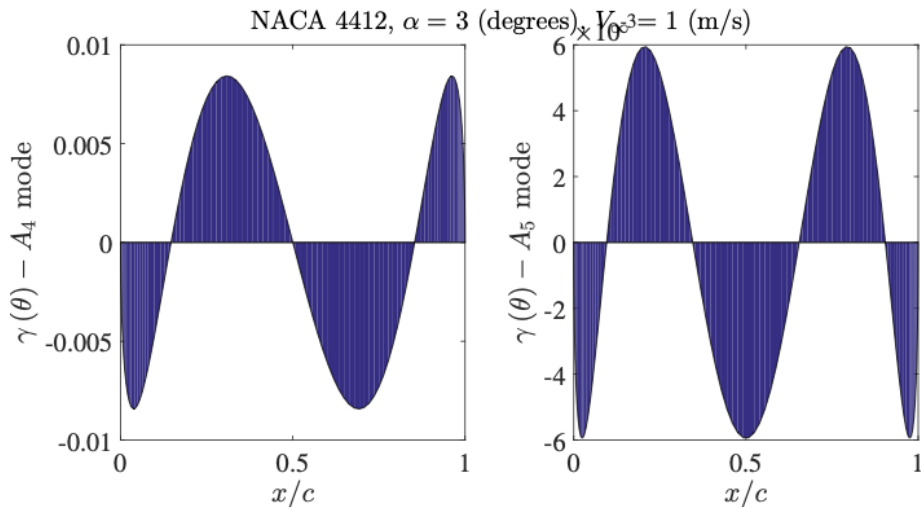


Solution for NACA 2412

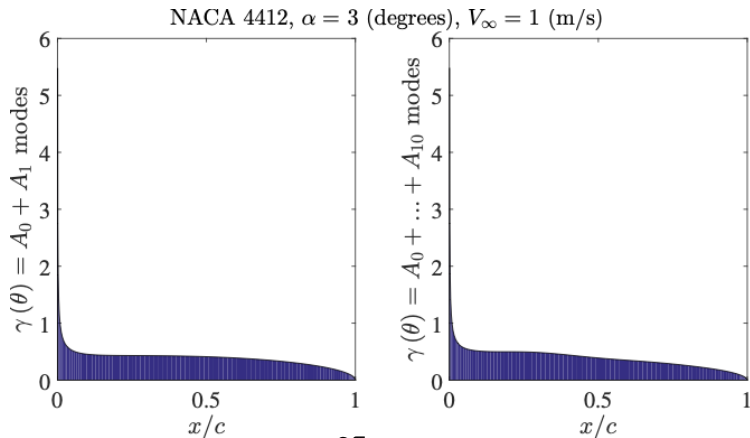
NACA 4412, $\alpha = 3$ (degrees), $V_\infty = 1$ (m/s)



Solution for NACA 2412

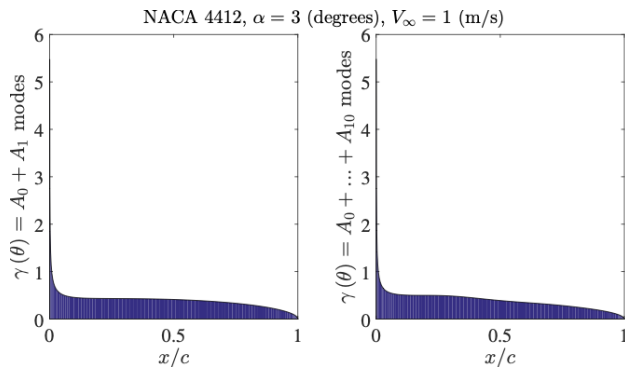


Solution for NACA 2412



$$\Gamma = \frac{c}{2} \int_0^\pi \gamma(\theta) \sin \theta \, d\theta$$

Solution for NACA 2412



At TE, $\frac{x}{c} = 1 \rightarrow \theta = \pi$, $\gamma(\text{TE}) = \gamma(c) = 0$, Kutta condn. is met (4.52)

Thin airfoil theory - derived quantities

Lift coefficient:

$$\Gamma = \int_0^c \gamma(\xi) d\xi = \frac{c}{2} \int_0^\pi \gamma(\theta) \sin \theta d\theta \quad (4.53)$$

$$= cU_\infty \left[A_0 \underbrace{\int_0^\pi (1 + \cos \theta) d\theta}_{=\pi} + \sum_{n=1}^{\infty} A_n \underbrace{\int_0^\pi \sin \theta \sin(n\theta) d\theta}_{\begin{cases} = 0, & \text{if } n \neq 1 \\ = \frac{\pi}{2}, & \text{if } n = 1 \end{cases}} \right] \quad (4.54)$$

$$= cU_\infty \left[\pi A_0 + \frac{\pi}{2} A_1 \right] \quad (4.55)$$

Thin airfoil theory - derived quantities

Lift coefficient:

$$l = \rho U_{\infty} \Gamma, \quad C_l = \frac{l}{\frac{1}{2} \rho U_{\infty}^2 c}$$

$$\text{Therefore, } C_l = 2\pi \left(A_0 + \frac{A_1}{2} \right) \quad (4.56)$$

$$= 2\pi \left[\alpha + \underbrace{\frac{1}{\pi} \int_0^{\pi} \frac{dz}{dx} (\cos \theta - 1) d\theta}_{=-\alpha_{0l}, \text{ zero-lift angle of attack}} \right] \quad (4.57)$$

$$= 2\pi (\alpha - \alpha_{0l}) \quad (\text{TEq. (6.25)}) \quad (4.58)$$

$$\frac{dC_l}{d\alpha} = 2\pi \quad (4.59)$$

Thin airfoil theory - derived quantities

Lift coefficient - symmetric airfoil:

$$C_l = 2\pi \left[\alpha + \underbrace{\frac{1}{\pi} \int_0^\pi \frac{dz}{dx} (\cos\theta - 1) d\theta}_{=-\alpha_{0l}, \text{ zero-lift angle of attack}} \right] = 2\pi \left(A_0 + \frac{A_1}{2} \right)$$

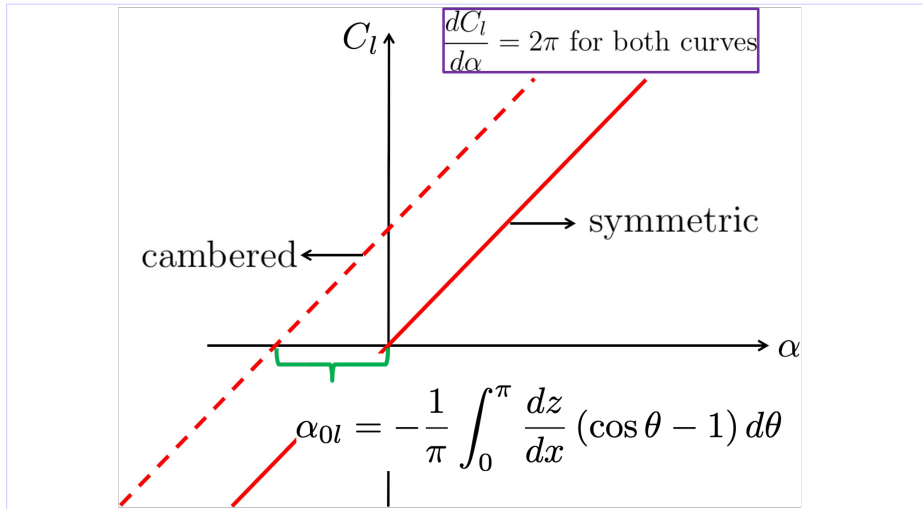
In the case of a symmetric airfoil, $z = 0 \rightarrow dz/dx = 0$. Therefore, for a symmetric airfoil,

$$A_0 = \alpha \text{ \& } A_n = 0, \quad n \geq 1 \quad (4.60)$$

$$\alpha_{0l} = 0 \quad (4.61)$$

$$C_l = 2\pi\alpha \quad (4.62)$$

Thin airfoil theory - coefficients



Thin airfoil theory - derived quantities

Moment coefficient:

$$m_0 = - \int_0^c \xi dl = -\rho U_\infty \int_0^c \xi \gamma(\xi) d\xi \quad (4.63)$$

$$C_{m0} = - \left[\frac{C_l}{4} + \frac{\pi}{4} (A_1 - A_2) \right] \quad (4.64)$$

$$= -\frac{\pi}{2} \left(A_0 + A_1 - \frac{A_2}{2} \right) \quad (4.65)$$

Symmetric airfoil, $A_0 = \alpha$, $A_1 = A_2 = 0$. Therefore in the case of a symmetric airfoil

$$C_{m0} = -\frac{\pi}{2} (\alpha) = -\frac{2\pi\alpha}{4} = -\frac{C_l}{4} \quad (4.66)$$

Thin airfoil theory - derived quantities

Center of pressure:

$$x_{cp} = c \frac{-C_{m0}}{C_l} \quad (4.67)$$

$$\frac{x_{cp}}{c} = \frac{1}{4} \left[1 + \frac{\pi}{C_l} (A_1 - A_2) \right] \quad (\text{TEq. (6.27)}) \quad (4.68)$$

Note: Here the location of x_{cp} depends on C_l which in turn is a function of α . Therefore the location of the center of pressure changes with the angle of attack α . However, in the case of the symmetric airfoil,

$$\text{Symmetric a/f} \rightarrow C_{m0} = -\frac{C_l}{4} \rightarrow x_{cp} = \frac{c}{4} \quad (4.69)$$

Therefore, in the case of a symmetric airfoil, the center of pressure is independent of the angle of attack

Thin airfoil theory - derived quantities

Aerodynamic center: Use transfer for moments i.e., eq. (4.27)

$$m_x = xl + m_0$$

$$C_{mx} = \frac{x}{c} C_l + c_{m0} \quad (4.70)$$

$$\text{choose } x = \frac{c}{4} \rightarrow C_{mc/4} = \frac{C_l}{4} + C_{m0} \quad (4.71)$$

$$C_{mc/4} = \frac{\pi}{4} (A_2 - A_1) \quad (\text{TEq. (6.29)}) \quad (4.72)$$

$$C_{mc/4} = \frac{\pi}{4} (A_2 - A_1), \text{ depends on camber but not } \alpha \quad (4.73)$$

$$\text{Therefore, } x_{ac} = \frac{c}{4} \text{ \& } C_{mac} = C_{mc/4} \quad (4.74)$$

Note, for a symmetric airfoil

$$C_{mac} = C_{mc/4} = C_{mx_{cp}} = 0 \quad (4.75)$$

Thin airfoil theory - Summary

$$z(x) \neq 0$$

$$\gamma(\theta) = 2U_\infty \left[A_0 \frac{(1 + \cos \theta)}{\sin \theta} + \sum_{n=1}^{\infty} A_n \sin(n\theta) \right]$$

$$C_l = 2\pi(\alpha - \alpha_{ZL}) = 2\pi \left(A_0 + \frac{A_1}{2} \right), \quad \frac{dC_l}{d\alpha} = 2\pi$$

$$C_{m0} = - \left[\frac{C_l}{4} + \frac{\pi}{4} (A_1 - A_2) \right]$$

$$\frac{x_{cp}}{c} = \frac{1}{4} \left[1 + \frac{\pi}{C_l} (A_1 - A_2) \right]$$

$$\frac{x_{ac}}{c} = \frac{1}{4}$$

$$C_{mac} = C_{mc/4} = \frac{\pi}{4} (A_2 - A_1)$$

$$C_d = 0$$

$$\text{Symmetric airfoil} \rightarrow A_0 = \alpha, \quad A_{n \geq 1} = 0$$

Thin airfoil theory - Is it any good?

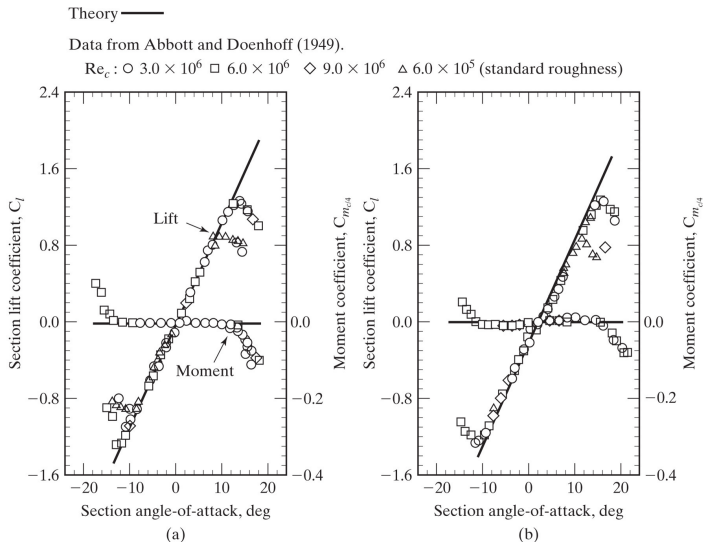


Figure 4.17: Solid lines are from thin airfoil theory. Fig 6.6 from AE-BC

Thin airfoil theory - Is it any good?

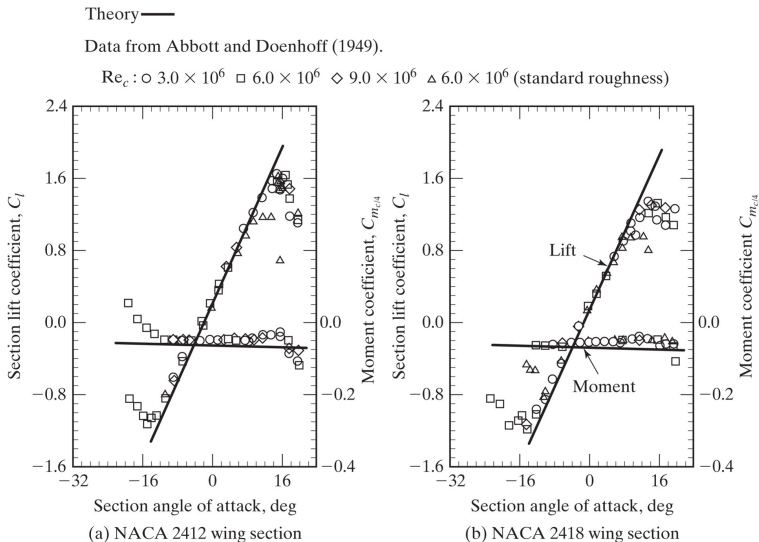
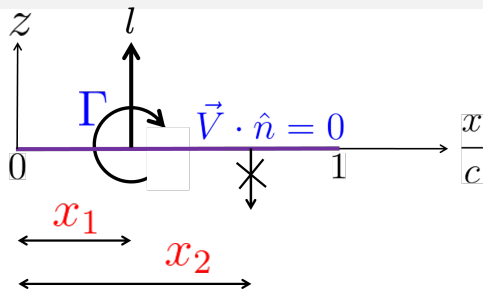


Figure 4.18: Solid lines are from thin airfoil theory. Fig 6.8 from AE-BC

Lumped vortex model



Instead of using a vortex sheet as in thin airfoil theory we lump all the vorticity into several panels along the camber line. For a start let's consider a single panel. This is also referred to sometimes as the lumped vortex model. let us place this lumped vortex at a distance x_1 from the leading edge. Since there is a single lumped vortex we can satisfy no-flow through the surface $\vec{V} \cdot \hat{n} = 0$ only at the one point. We satisfy no-flow through the surface at some point x_2 from the leading edge

Lumped vortex model

According to thin airfoil theory (TAT), for a symmetric airfoil $C_l = 2\pi\alpha$ and $C_{m0} = -C_l/4$. So even if we represent this symmetric flat plate airfoil as a lumped vortex we should get the same lift and moment as derived from thin airfoil theory. So the locations of these points x_1 and x_2 are determined by this thin airfoil theory result.

$$\text{At } x_2 \rightarrow \vec{V} \cdot \hat{n} = 0 \quad (4.76)$$

$$\rightarrow U_\infty \sin \alpha - \frac{\Gamma}{2\pi(x_2 - x_1)} = 0 \quad (4.77)$$

$$\rightarrow x_2 - x_1 = \frac{\Gamma}{2\pi U_\infty \sin \alpha} \quad (4.78)$$

$$\text{Small angle approx. } \sin \alpha \approx \alpha \quad (4.79)$$

$$\rightarrow x_2 - x_1 = \frac{\Gamma}{2\pi U_\infty \alpha} \quad (4.80)$$

Lumped vortex model

$$c_{m0} = \frac{-x_1 l}{q_\infty c c} \quad (4.81)$$

$$= -\frac{x_1}{c} C_l = -\frac{C_l}{4} \quad (\text{from TAT}) \quad (4.82)$$

$$\rightarrow x_1 = \frac{c}{4} \quad (4.83)$$

$$(4.84)$$

substitute in eq. 4.80 we get

$$x_2 = x_1 + \frac{\Gamma}{2\pi U_\infty \alpha} \quad (4.85)$$

$$\text{K-J theorem, } l = \rho U_\infty \Gamma \quad (4.86)$$

$$\rightarrow C_l = \frac{2\rho U_\infty \Gamma}{\rho U_\infty^2 c} = \frac{2\Gamma}{U_\infty c} \quad (4.87)$$

Lumped vortex model

$$\rightarrow \Gamma = \frac{C_l U_\infty c}{2} \quad (4.88)$$

$$\text{therefore, } x_2 = x_1 + \frac{C_l U_\infty c}{4\pi U_\infty \alpha} = \frac{C_l c}{4\pi \alpha} \quad (4.89)$$

$$\text{from TAT, } C_l = 2\pi\alpha, \rightarrow x_2 = x_1 + \frac{c}{2} \quad (4.90)$$

$$x_2 = \frac{3c}{4} \text{ as, } x_1 = \frac{c}{4} \quad (4.91)$$

Therefore, for a lumped vortex model we place the vortex at quarter chord ($c/4$) of that panel and satisfy no flow through the surface at three-quarter chord ($3c/4$) of that panel

Method of Images

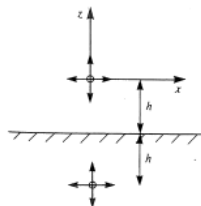


Figure 6.20 Image of source in plane wall.

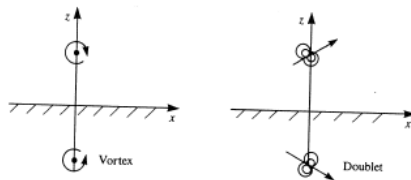


Figure 6.21 Image of vortex and doublet in plane wall.

Figure 4.19: From Low-Speed Aerodynamics Katz & Plotkin

Method of Images

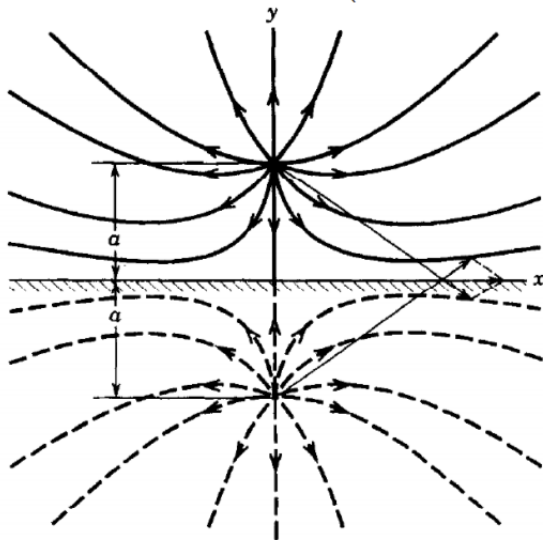


Figure 4.20: From FA-KC

Method of Images

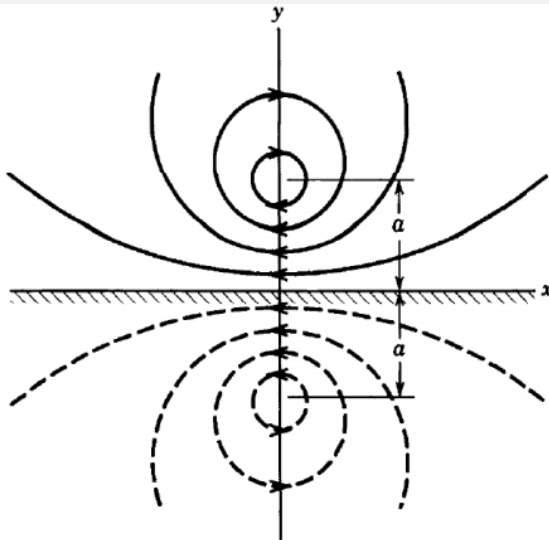


Figure 4.21: From FA-KC

Method of Images

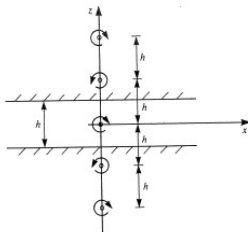


Figure 6.23 Image of clockwise vortex midway between parallel walls.

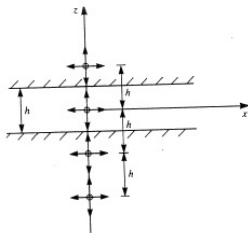


Figure 6.22 Image of source midway between parallel walls.

Figure 4.22: From Low-Speed Aerodynamics Katz & Plotkin

Method of Images

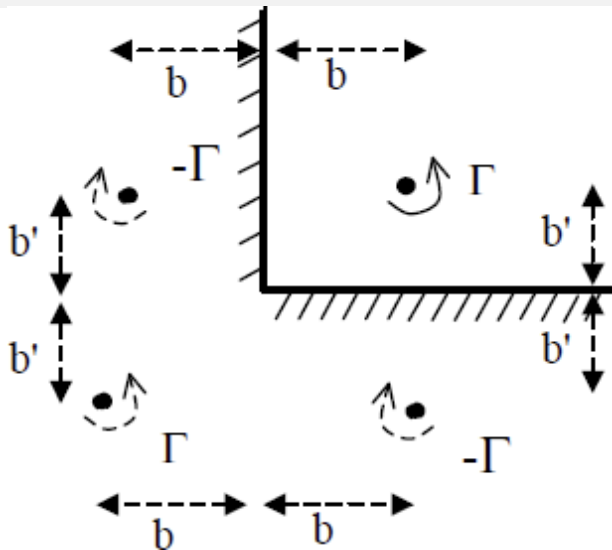
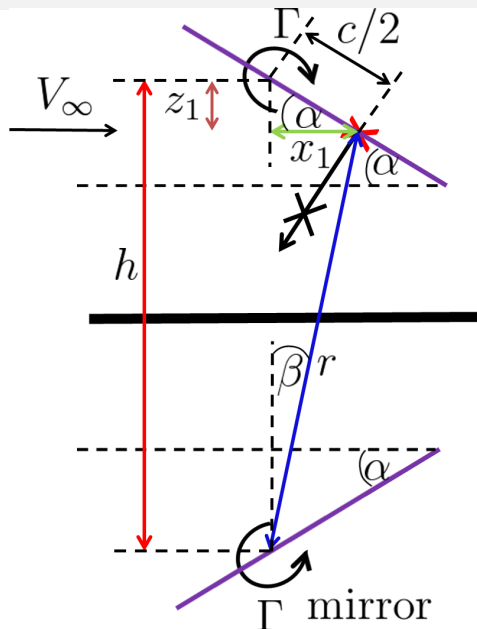


Figure 4.23: From MIT open course ware

Worked out example



$$\sin \beta = \frac{x_1}{r}$$

$$r = \sqrt{(h - z_1)^2 + x_1^2}$$

$$x_1 = \frac{c}{2} \cos \alpha$$

$$z_1 = \frac{c}{2} \sin \alpha$$

Component of mirror at x

$$= \frac{\Gamma}{2\pi r} \sin(\pi - \beta)$$

Worked out example

10/31/15 12:28 PM C...\TATLumpedVor.m 1 of 2

```
clc;
clear all;
close all;

alpha=5; % in degrees
Vinf=1; % in ms^-1
h=[0.35 0.4 0.5 0.75 1 1.5 2 3 5 7 10 20 50 100 200]; % in m

alphar=alpha/180*pi; % convert alpha to radians

% airfoil alone
C_l=2*pi*alphar

% in ground effect
h_by_c=h.*0;
C_l_ground=h.*0;
for i=1:length(h)
    c=1;
    x1=c/2*cos(alphar);
    z1=c/2*sin(alphar);
    r=sqrt((h(i)-z1)^2+x1^2);
    beta=asin(x1/r);

    Gamma=Vinf*alphar*2*pi/(2/c-1/r*sin(pi-beta));
    C_l_ground(i)=2*Gamma/Vinf/c;
    h_by_c(i)=h(i)/c;
end

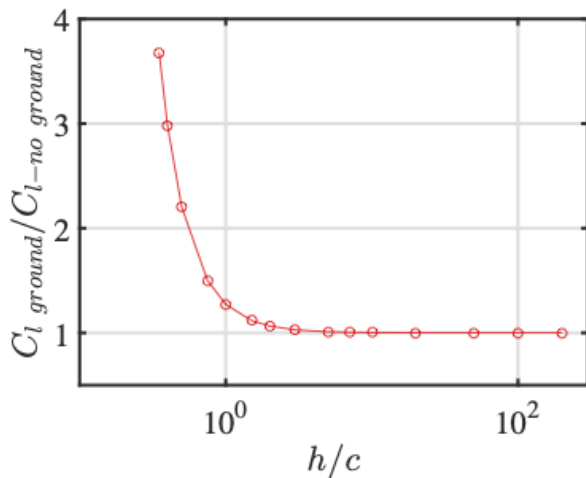
set(
(0,'DefaultTextInterpreter','latex','DefaultAxesLineWidth',✓
2,...
'DefaultAxesFontSize',✓
20,'DefaultFigureInvertHardCopy',...
```

Worked out example

10/31/15 12:28 PM C...\TATLumpedVor.m 2 of 2

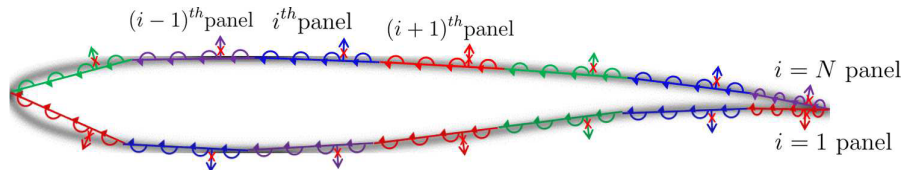
```
'on', 'DefaultAxesFontName', 'Times')  
  
figure  
semilogx(h_by_c, C_l_ground./C_l, 'o-r')  
grid;  
xlabel('$h/c$')  
ylabel('$C_{l-ground}/C_{l-no-ground}$')  
axis([0.1 300 0.5 4])
```

Worked out example



Note: The x ordinate is in logarithmic scale

Vortex panel method



1. Enforce $\vec{V} \cdot \hat{n} = 0$ at N control (collocation) points
2. Enforce the Kutta condition at the trailing edge
 $\rightarrow \gamma(TE) = 0 \rightarrow \gamma_1 = -\gamma_N$
3. This gives us $N + 1$ equations for N ($\gamma_1 - \gamma_N$) unknowns $\rightarrow$ over determined set of equations
 - 3.1 Ignore $\vec{V} \cdot \hat{n} = 0$ at one control point
 - 3.2 Use a least squares approach
 - 3.3 Use another type of vortex panel method

Vortex panel method

Enforce $\vec{V} \cdot \hat{n} = 0$ at control point i :

$$(U_\infty \cos \alpha + \tilde{u}) n_x(i) + (U_\infty \sin \alpha + \tilde{w}) n_z(i) = 0 \quad (4.92)$$

$$\tilde{u} n_x(i) + \tilde{w} n_y(i) = -U_\infty [\cos \alpha n_x(i) + \sin \alpha n_y(i)] \quad (4.93)$$

$$\tilde{u} = \sum_{j=1}^N \{u \text{ velocity induced by panel } j \text{ at control point } i\} \quad (4.94)$$

$$\tilde{w} = \sum_{j=1}^N \{w \text{ velocity induced by panel } j \text{ at control point } i\} \quad (4.95)$$

$$\tilde{u} n_x(i) + \tilde{w} n_y(i) = \sum_{j=1}^N \frac{\gamma_j}{2\pi} J_{ij} \quad (4.96)$$

$$\sum_{j=1}^N \frac{\gamma_j}{2\pi} J_{ij} = -U_\infty [\cos \alpha n_x(i) + \sin \alpha n_y(i)] \quad (4.97)$$

Vortex panel method

Another commonly used panel method is the Smith Hess Panel Method. I have coded it in MATLAB and you will use the code as part of your homework.